Chapter 14 Notes

Section 1

Notes

Let K and K' be classes, K be transitive, $K \subseteq K'$, and $\phi(x_1, ..., x_n)$ a formula whose constants are all in K. Note that, even if we assume $\phi(a_1, ..., a_n)$ is true over K iff it is true over K' (for all constants $a_1, ..., a_n$ in K), we *cannot* prove that $(\exists x)(\phi(x, x_1, ..., x_n))$ is such that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over K iff it is true over K' (for all constants $a_1, ..., a_n$ in K); call any ϕ *good* if this property does hold for it. For, let $a_1, ..., a_n$ be constants in K. On the one hand, suppose that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over K. Then for some $a \in K$, $\phi(a, a_1, ..., a_n)$ is true over K, and thus by assumption, $\phi(a, a_1, ..., a_n)$ is true over K', hence $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over K'. However, on the other hand, suppose that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over K'. Then for some $a \in K'$, $\phi(a, a_1, ..., a_n)$ is true over K'. However, unlike in Exercise 1.1 (a), we *cannot* conclude that $a \in K$, meaning it does not necessarily follow that $\phi(a, a_1, ..., a_n)$ is true over K (and thus it does not necessarily follow that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over K.

Using the same definitions as in the preceeding paragraph, note the following. Suppose $\phi(x_1, ..., x_n)$ is only such that if $\phi(a_1, ..., a_n)$ is true over K (for constants $a_1, ..., a_n$ in K), then $\phi(a_1, ..., a_n)$ is true over K'. It certainly does *not* follow that $\sim\phi$ is good (or even satisfies the weaker property if if $\sim\phi(a_1, ..., a_n)$ is true over K, then it is true over K'). For, if $\sim\phi(a_1, ..., a_n)$ is true over K, then $\phi(a_1, ..., a_n)$ is false over K. But because we do not have that ϕ is good, we cannot conclude that $\phi(a_1, ..., a_n)$ is false over K' (now, if $\sim\phi(a_1, ..., a_n)$ was true over K', then $\phi(a_1, ..., a_n)$ would be false over K', and then by assumption, $\phi(a_1, ..., a_n)$ would also need to be false over K, hence $\sim\phi(a_1, ..., a_n)$ would be true over K).

The above paragraph explains why Σ-formulas do not include negations (unless the subformlula the negation is the main operator of is itself a Σ_0 formula).

Proposition 1.1

Note that part (3) of this proposition follows in a similar way as part (3) of Exercise 1.1 (b) is proven below.

Exercise 1.1 (a)

Suppose K and K' are classes, such that K is transitive, $K \subseteq K'$, $a_1, ..., a_n$ are elements of K, and $\phi(x_1, ..., x_n)$ is a formula whose constants are all in K. Suppose also that $\phi(x_1, ..., x_n)$ is Σ_0. We are to show that $\phi(a_1, ..., a_n)$ is true over K iff it is true over K'. We prove this by induction on the degree of $\phi(x_1, ..., x_n)$.

Base Case: If $\phi(x_1, ..., x_n)$ is of the form $x \in y$, then $a_1 \in a_2$ is true over K iff it is true over K' (as a_1 and a_2 are both in K', as $K \subseteq K'$).

If the result holds for some ϕ, then obviously it also holds for $\sim\phi$.

If the result holds for ϕ and ψ, then obviously it also holds for $\phi \wedge \psi$.

Suppose the result holds for some $\phi(x, y, x_1, ..., x_n)$ (whose free variables are all in the set $\{x, y, x_1, ..., x_n\}$). We must show that the result also holds for

$$(\exists x \in y)(\phi(x, y, x_1, ..., x_n))$$

Take any constants $a, b_1, ..., b_n$ of K. We must show that $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over K iff it is true over K'. On the one hand, suppose that it is true over K. Then for some $b \in a$, $\phi(b, a, b_1, ..., b_n)$ is true over K. But the constants $b, a, b_1, ..., b_n$ are all in K, meaning by assumption, $\phi(b, a, b_1, ..., b_n)$ is true over K', and thus $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over K'. On the other hand, suppose that $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over K'. Then for some $b \in a$, $\phi(b, a, b_1, ..., b_n)$ is true over K'. But K is transitive, and $a \in K$, meaning $b \in K$. Thus the constants $b, a, b_1, ..., b_n$ are all in K, meaning, by assumption, $\phi(b, a, b_1, ..., b_n)$ is true over K, and thus $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over K.

Exercise 1.1 (b)

Suppose K and K' are classes, such that K is transitive, $K \subseteq K'$, $a_1, ..., a_n$ are elements of K, and $\phi(x_1, ..., x_n)$ is a formula whose constants are all in K. Suppose also that $\phi(x_1, ..., x_n)$ is Σ. We are to show that $\phi(a_1, ..., a_n)$ is true over K iff it is true over K'. Call a formula ϕ good if it satisfies this result. We prove this by induction on the degree of $\phi(x_1, ..., x_n)$. In fact, we prove:

(1) Every Σ_0 formula is good.

(2) If ϕ, ψ are good, then so are $\phi \wedge \psi$ and $\phi \vee \psi$.

(3) If ϕ is good, so is $(\exists x)\phi$.

(4) If ϕ is good, so are $(\forall x \in y)\phi$ and $(\exists x \in y)\phi$.

Proof. (1) This follows by Exercise 1.1 (a).

(2) Obvious.

(3) Suppose $\phi(x, x_1, ..., x_n)$ is good (where all free variables of ϕ are in the set $\{x, x_1, ..., x_n\}$). We must show that $(\exists x)\phi(x, x_1, ..., x_n)$ is good. So suppose $(\exists x)\phi(x, a_1, ..., a_n)$ was true over K. Then, for some $a \in K$, $\phi(a, a_1, ..., a_n)$ is true over K, hence by assumption, $\phi(a, a_1, ..., a_n)$ is true over K', meaning $(\exists x)\phi(x, a_1, ..., a_n)$ is true over K'.

(4) Suppose $\phi(x, y, z_1, ..., z_n)$ is good. We wish to show that $(\forall x \in y)\phi(x, y, z_1, ..., z_n)$ is also good. Well, let $a, b_1, ..., b_n$ be constants in K. Suppose $(\forall x \in a)\phi(x, a, b_1, ..., b_n)$ is true over K. Because K is transitive, it follows that $\phi(c, a, b_1, ..., b_n)$ is true over K for all elements $c \in a$. Then, by the assumption that ϕ is good, it follows that $(\forall x \in a)\phi(x, a, b_1, ..., b_n)$ is true over K'.

As for the formula $(\exists x \in y)\phi$, this is $(\exists x)(x \in y \wedge \phi)$, which is good by parts (1), (2), and (3) of this exercise. $\square$